Jaron Pressman-Cyna

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Education

Queen's University Kingston, ON
Electrical Engineering / Engineering Physics, GPA: 4.07/4.3 *Expected May 2028*

- **Relevant Coursework:** Digital Systems, Electronics I-III, Power Electronics, Computational Eng. Phys (Python), Solid State Devices, Signals and Systems, Engineering-Design I-IV

Technical Projects

5-Stage RV32I SoC (FOC Accelerator) | *SystemVerilog, Computer Organization* May. 2026 – Aug. 2026

- Designed a 5-stage pipelined RV32I CPU in SystemVerilog, achieving **192.42 MHz Fmax** when synthesized on an Intel Agilex 3 FPGA **with 1.12 CPI**.
- Integrated custom MMIO coprocessors, including a 16-stage pipelined CORDIC engine, cutting Field-Oriented Control (FOC) loop calculation latency by **5x (3.1 μs down to 624 ns)** compared to C firmware baseline.
- Verified microarchitectural edge cases using a **20,000-line constrained-random test (CRT)** suite, achieving 100% functional coverage across all pipeline hazards.
- Implemented full data forwarding (EX/MEM-to-EX) and hazard detection units, resolving RAW dependencies and load-use stalls with hardware-enforced pipeline bubbles.
- Developed bare-metal C drivers and control firmware interfacing directly with custom MMIO registers, **achieving a** $<$ 900**-byte** memory footprint by eliminating C runtime overhead.

Maximum Power Point Tracking CubeSat PCB | *Mixed Signal, Power Electronics* Jan. 2026 – Aug. 2026

- Architected a 6-layer mixed signal EPS PCB in KiCad delivering regulated 3.3V/5V rails, **simulating** $<$ 15 **$mV_{p\text{-}p}$ ripple** in LTSpice by using real board parasitic inductance and capacitance.
- Engineered a discrete hardware watchdog using radiation-hardened D Flip-Flops to ensure deterministic recovery and autonomy via an ideal diode OR-ing circuit.
- Eliminated memory corruption and Single-Event Upset (SEU) bit-flips in the storage by utilizing SPI Magnetoresistive RAM (MRAM) for persistent telemetry tracking with unlimited write endurance.

Experience

SNO+ Experiment May. 2026 – Aug. 2026
Undergraduate Researcher (USSRF Grant) *Kingston, ON*

- Salvaged **5+ months of unviable experimental data** by developing an automated calibration pipeline that leveraged a Polonium-210 localized standard candle to dynamically correct shifting optical detector properties.
- Accelerated event processing throughput by **16x on 2,000,000+ events** via C++ branch-level masking, minimizing memory footprint and I/O overhead across multi-terabyte datasets.
- Parallelized 3D spatial fitting routines across hundreds of detector runs via automated SLURM batch scripts, scaling across 20 parallel compute nodes to reduce processing time from **30+ hours to under 2**.

Queen's Space Engineering Team Sept. 2024 – Present
Satellite Co-Chief Technical Officer & prev. Rover Firmware Team Lead *Kingston, ON*

- Enabled a transition **from I^2C to CAN bus** for inter-board communication, eliminating noise vulnerabilities and establishing fault-tolerant telemetry.
- Standardized a PC-104 electrical interconnect architecture across all 6 subsystem PCBs, eliminating pinout incompatibilities and unifying power/bus routing for a 100-member satellite team.
- Engineered STM32F4 FreeRTOS firmware to concurrently control 8+ actuators and scientific sensors, contributing to the 88/100 Science Mission score at URC 2026.
- Modernized engineering infrastructure by deploying Git-based documentation (Sphinx) and agile issue tracking, driving 100% adoption across 7 subteams to streamline design reviews.

Technical Skills

Languages: C, C++, Python, SystemVerilog, Verilog, VHDL
Embedded Systems & Hardware: Assembly (RISC-V, ARM), RTL Design, STM32, FreeRTOS, ESP32, FPGA, CAN Bus, SPI/I2C/UART, PCB Design (Altium, KiCad), Oscilloscope, Signal Conditioning, Synchronous Rectification
Software & HPC: SLURM Job Scheduling, CMake, Git, Parallelization, GitHub, Linux/WSL, PowerShell, ModelSim, Quartus, CUDA, Ansys, LTspice, SolidWorks, PyTorch, VS Code